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Full-length Article

Bifidobacterium breve CCFM1025 attenuates major depression disorder via regulating gut microbiome and tryptophan metabolism: A randomized clinical trial

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A B S T R A C T

Objective: Psychobiotics, as a novel class of probiotics mainly acting on the gut-brain axis, have shown promising prospects in treating psychiatric disorders. *Bifidobacterium breve* CCFM1025 was validated to have an antidepressant-like effect in mice. This study aims to assess its psychotropic potential in managing major depression disorder (MDD) and unravel the underlying mechanisms.
Methods: Clinical Trial Registration: https://www.chictr.org.cn/index.aspx (identifier: NO. ChiCTR2100046321). Patients (n = 45) diagnosed with MDD were randomly assigned to the *Placebo* (n = 25) and *CCFM1025* (n = 20) groups. The freeze-dried CCFM1025 in a dose of viable bacteria of 10^{10} CFU was given to MDD patients daily for four weeks, while the placebo group was given maltodextrin. Changes from baseline in psychometric and gastrointestinal symptoms were evaluated using Hamilton Depression Rating scale-24 Items (HDRS-24), Montgomery-Asberg Depression Rating Scale (MADRS), Brief Psychiatric Rating Scale (BPRS), and Gastrointestinal Symptom Rating Scale (GSRS). Serum measures were also determined, i.e., cortisol, TNF-α, and IL-β. Serotonin turnover in the circulation, gut microbiome composition, and tryptophan metabolites were further investigated for clarifying the probiotics' mechanisms of action.
Results: CCFM1025 showed a better antidepressant-like effect than placebo, based on the HDRS-24 (placebo: $M = 6.44$, $SD = 5.44$; CCFM1025: $M = 10.40$, $SD = 6.85$; $t(43) = 2.163$, $P = 0.036$, $d = 0.640$) and MADRS (placebo: $M = 4.92$, $SD = 7.15$; CCFM1025: $M = 9.60$, $SD = 7.37$; $t(43) = 2.152$, $P = 0.037$, $d = 0.645$) evaluation. The factor analysis of BPRS and GSRS suggested that patients' emotional and gastrointestinal problems may be affected by the serotonergic system. Specifically, CCFM1025 could significantly and to a larger extend reduce the serum serotonin turnover compared with the placebo (placebo: $M = -0.01$, $SD = 0.41$; CCFM1025: $M = 0.27$, $SD = 0.40$; $t(43) = 2.267$, $P = 0.029$, $d = 0.681$). It may be due to changes in gut microbiome and gut tryptophan metabolism under the probiotic treatment, such as changes in alpha diversity, tryptophan, and indoles derivatives.
Conclusion: B. breve CCFM1025 is a promising candidate psychobiotic strain that attenuates depression and associated gastrointestinal disorders. The mechanisms may be relevant to the changes in the gut microbiome and tryptophan metabolism. These findings support the future clinical applications of psychobiotics in the treatment of psychiatric disorders.

1. Introduction

Major depression disorder (MDD) is a heterogeneous disease caused by both genetic and environmental factors. Patients who suffer from depression represent multiple psychometric symptoms such as persistent low mood, anxiety and panic, sleep disorders, and sexual dysfunction. Severely depressed patients show significant tendencies of self-injury or suicide, which makes MDD one of the leading mortality risks among human diseases. During the past year, the spreading of the Severe Acute Respiratory Syndrome Coronavirus (COVID-19) pandemic seems to worsen global mental wellbeing situations (Organization, 2020). The widespread anxiety, social isolation, financial stress, and traumatic memories dramatically increased people's susceptibility to MDD (Rogers et al., 2020). The use of traditional psychotropic drugs, such as Selective Serotonin Reuptake Inhibitor (SSRIs) (El Mansari et al., 2005; Thase et al., 2001; Trivedi et al., 2006), is often reported to be

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ineffective, and it could post endocrine and gastrointestinal side effects. A novel type of therapy is urgently and almost inevitably needed.

With increasing knowledge of the gut microbiome's role in human health, the therapies targeting the gut microbiome bring hope for preventing and treating MDD, such as consuming prebiotics or probiotics. The microbiome-oriented supplements showed a less aggressive effect on the body and apparent psychobiotic potentials (Ansari et al., 2020; Liu et al., 2019). In 2013, the term "psychobiotics" was first defined as probiotics that benefit the health of patients suffering from psychiatric illnesses (Dinan et al., 2013). In recent years, the interaction between psychobiotics and the gut-brain axis and their underlying mechanisms have gained increasing attention from doctors and scientists (Cepeda et al., 2017; Mörkl et al., 2020). Psychobiotics' antidepressant-like effects are widely reported in animal studies and are increasingly validated in human trials when used as adjuvant psychotropic drugs (Rudzki et al., 2019; Slykerman et al., 2017). Intake of psychobiotics in daily life also showed positive effects on managing stress and preventing mental disorders (Allen et al., 2016; Marotta et al., 2019; Pinto-Sanchez et al., 2017). However, the present clinical trial provided minimal knowledge on the gut microbiome and metabolome features under psychobiotic treatment.

RIN14B cells are derived from the islet tumor cells of rats due to their high expression of key markers of enterochromaffin cells, such as chromogranin A, synaptophysin, vesicular monoamine transporter 1 (VMAT1), tryptophan hydroxylase 1 (TPH1), and transient receptor potential ankyrin subtype 1 (TRPA1). They are widely used as a putative enterochromaffin cell model (Nozawa et al., 2009; Yano et al., 2015). In our previous research, the RIN14B cell was creatively used to screen the psychobiotics that could enhance gut-derived serotonin (or its precursors) production (Tian et al., 2021b), and these strains may then affect the brain's function via the microbiome-gut-brain axis (Tian et al., 2019a). *Bifidobacterium breve* CCFM1025, as one of the most effective strains, was chosen to further validate its antidepressant-like effect in chronic stress-induced depressed mice (Wu et al., 2020). Specifically, using multiple techniques involving microbiology, neuropharmacology, and omics analysis, the underlying mechanisms of the CCFM1025′s antidepressant-like potential were unraveled from the perspective of the recipients' neurobiological changes and the bacteria *per se*, such as their metabolomic biomarkers and colonization capacity (Tian et al., 2021a). This study aims to investigate the CCFM1025′s effectiveness in managing MDD. The gut microbial and metabolomic effect of CCFM1025 will also be addressed.

2. Material and Methods

2.1. Ethical approvement

The trial was approved by the Research Ethics Committee of the Tinghu People's Hospital (Yancheng, China; Approval Code: ET2020076) and retrospectively registered in the Chinese Clinical Trial Registry (NO. ChiCTR2100046321). All patients provided written informed consent before enrollment. A multiple-probiotics intervention procedure was approved and registered simultaneously with independent controls from this study. It focuses on another story, and the investigation is still in progress and thus is not reported here. The TNF-α concentration in serum and tryptophan metabolites in the feces were externally determined to investigate the probiotics' potential mechanisms (using the redundant samples within the registered study design). However, the indicators were not included in the registered protocols, which has to be pointed out here.

2.2. Sample size estimation

The prior power analysis was calculated using G*Power 3.1. The estimated effect size was obtained from a published study that used probiotics to treat MDD by comparing the mean difference of the Hamilton Depression Rating Scale score between the probiotics- and placebo-treated groups (Majeed et al., 2018). To detect a power of 0.8 for a *t*-test analysis, groups of fifteen participants are required to achieve an effect sized $d = 0.936$ at $\alpha = 0.05$ (detailed analysis procedure was shown in Supplemental Fig.S1. We assume a dropping rate of 25% during the experiment, so $15/(1–0.25) = 20$ participants per group are needed.

2.3. Study design

The study was a placebo-controlled, double-blind Randomized Controlled Trial (RCT). The CONSORT flow diagram is shown in Fig. 1. Subjects were blinded and given intervention products dispensed independently using randomization code (using random number table (Suresh, 2011)) by an authorized investigator. MDD patients over 18 years were included, with no restrictions on taking antidepressant drugs, such as SSRIs. Patients diagnosed with refractory depression and with severe suicide and self-injury tendencies were excluded. Patients with other severe physical and neurological diseases (such as HIV, heart diseases, schizophrenia, bipolar disorder, and neurodegenerative diseases, epilepsy) and pregnant or breast-feeding females were excluded.

Patients who scored over 14 in HDRS-24 were identified to have MDD. Based on the doctors' suggestions, several patients who did not fulfill this standard but had a severe sleeping disorder or MDD tendency were also included in the cohort. Each patient's course of disease has been provided in Supplemental Table S1. Finally, forty-five MDD patients finished the trial (*Placebo*, n = 25, female rate 50%; *CCFM1025*, n = 20, female rate 70%). The demographic characteristics including age, sex, and antidepressant use were summarized in Table 1 (Supplemental Table S1 shows the demographic information of each participant).

Patients consumed a sachet containing freeze-dried *Bifidobacterium breve* CCFM1025 powder (with a total viable bacteria of 10^{10} CFU) or maltodextrin as the placebo for four weeks (from day 0 to day 28, please see the experimental schedule in Supplemental Fig.S2. There are no particular diet restrictions, except for avoiding taking antibiotics and probiotics simultaneously. The recruitment, intervention, and sample collection started on Oct 1st, 2020, and finished on Dec 31st, 2021. The subsequent lab test and data analysis were completed in July 2021. The sachets were developed by Shisheng Yisheng (Yangzhou), Co., Ltd. The packaging, color, and quality of the placebo sachet are the same as the probiotic sachet.

2.4. Depressive symptoms (primary outcome measures)

The primary outcome for this study was the mean change of the depression symptoms assessed with psychometric questionnaires, including Hamilton Depression Rating scale-24 Items (HDRS-24) (Henigsberg et al., 2012), Montgomery-Asberg Depression Rating Scale (MADRS) (Guizzaro et al., 2020), and Brief Psychiatric Rating Scale (BPRS) (Lachar et al., 1999). Notably, the BPRS items were classified into five aspects, including "Hostility", "Thinking disorder", "Activation", "Withdrawal-Retardation", and "Anxiety-Depression" (Lachar et al., 1999). The scoring was completed by the doctors when they were examining the patients.

2.5. Gastrointestinal symptoms (primary outcome measures)

Gastrointestinal Symptom Rating Scale (GSRS) includes 16 items in a multiple-choice format (Kulich et al., 2008). The items were classified into "Bowel dysfunction syndrome", "Indigestion syndrome", "Dyspeptic syndrome", and "Abdominal pain syndrome". The GSRS scores were generated from self-evaluation according to the feelings during the past week.

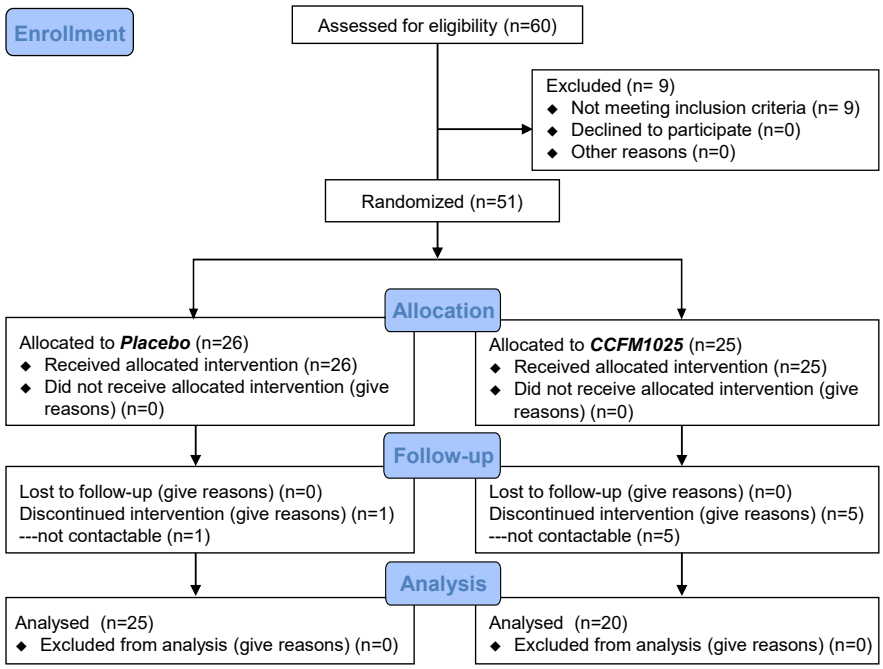

Fig. 1. Flow chart of study procedures.

Table 1
Demographic Information.

Variable		CCFM1025 (n = 20)	Placebo (n = 25)	Statistic Value	*P* value
Age, Mean ± SD		51.32 ± 16.11	48.15 ± 13.96	0.695	0.491 [a]
Sex, n (%)	Female	14 (70)	16 (64)	0.180	0.671 [b]
	Male	6 (30)	9 (36)		
Race, n (%)	Chinese, Han nationality	20 (100)	25 (100)	N/A	N/A
Medication before inclusion, n (%)		18 (90)	22 (88)	0.045	0.832 [b]
Medication type before inclusion, n (%)	Escitalopram	0 (0)	2 (8)	1.722	0.189 [b]
	Sertraline	0 (0)	1 (4)	0.839	0.359 [b]
	Paroxetine	0 (0)	1 (4)	0.839	0.359 [b]
	Oxazepam	3 (15)	4 (16)	0.016	0.900 [b]
	Olanzapine	4 (20)	4 (16)	0.101	0.751 [b]
	Fluvoxamine	1 (5)	0 (0)	1.197	0.274 [b]
	Duloxetine	4 (20)	3 (12)	0.506	0.477 [b]
	Lorazepam	2 (10)	2 (8)	0.045	0.832 [b]
	Fluoxetine	3 (15)	1 (4)	1.616	0.204 [b]
	Venlafaxine	0 (0)	2 (8)	1.722	0.189 [b]
	Traditional Chinese Medicine	1 (5)	2 (8)	0.178	0.673 [b]
	Unknown	6 (30)	7 (28)	0.010	0.919 [b]
Educational status, n (%)	Primary	3 (15)	3 (12)	2.532	0.469 [b]
	Secondary	5 (25)	12 (48)		
	Higher	5 (25)	4 (16)		
	Unknown	5 (25)	4 (16)		
Alcohol use,n (%)	Non-drinker	17 (85)	20 (80)	0.190	0.663 [b]
	Occasional drinker	3 (15)	5 (20)		
Smokers, n (%)	Non-smoker	18 (90)	23 (92)	0.815	0.055 [b]

n, number of participants
N/A, not available
[a] Statistical analysis by *t*-test
[b] Statistical analysis by chi square test

2.6. Serum measures (secondary outcome measures)

The serum 5-HT and 5-hydroxyindoleacetic acid (5-HIAA) were determined using the C18 column (150 mm × 4.6 mm, 5 μm) with the HPLC system (Waters 2695) and fluorescence detector (Waters 2475) as previously described (Tian et al., 2020; Tian et al., 2019b). The 5-HT and 5-HIAA stands were purchased from Sigma-Aldrich (#H9523 and #H9772, Shanghai, China). The 5-HT turnover was defined as the 5-HIAA: 5-HT ratio. Serum cortisol, TNF-α, and IL-β were measured using ELISA kits according to the manufacturers' instructions (Elabscience Co., Ltd., Wuhan, China).

2.7. Fecal microbiome analysis (primary outcome measures)

Fecal samples were collected from all subjects at the baseline and endpoint. The total DNA of bacteria in the fresh stool was extracted using the Fast DNA Stool Kit (MP Biomedicals, California, USA). 16S rRNA gene amplicon sequencing was performed using universal primers (341F/806R, for V3-V4 region), as previously described (Tian et al., 2021a). Raw data were processed using the QIIME2 software package, and the specialized bioinformatic analysis was performed using the online software of MicrobiomeAnalyst (https://www.microbiomeanalyst.ca), which included alpha diversity, beta diversity, core microbiome analysis, and random forest modelling (Chong et al., 2020; Dhariwal et al., 2017).

2.8. Gut tryptophan pathway-targeted metabolomic analysis

According to the previous report, the tryptophan metabolites were quantitatively determined with some modifications (González et al., 2011). Fresh feces samples were consecutively frozen and thawed using liquid nitrogen three times, followed by freeze-drying at −107℃ and 0.2 mbar pressure. About 50 mg freeze-dried fecal sample was dissolved in 900 μL of MeOH-H_2O (1:1), the mixture was then homogenized and centrifuged at 15000 × g for 10 min. The supernatant was collected and concentrated in a vacuum (45℃, 4 h). After another centrifuge at 15000

× g for 10 min, the supernatant was ready for ultra-high performance liquid chromatography-tandem mass spectrometry (UHPLC-MS/MS) determination for metabolites of tryptophan metabolism. The chromatographic conditions are described in *Supplemental 1.1*. The multiple reaction monitoring (MRM) transitions and the cone voltages and collision energies are shown in Table S1. The concentration of metabolites was quantified according to the external standard curve. Tryptophan pathway-targeted metabolomic analysis was carried out using the online software of MetaboAnalyst 5.0 (http://www.metaboanalyst.ca/) (Pang et al., 2021).

2.9. Statistical analysis

All data were checked for normality with the Shapiro-Wilk test before further analyses. The two-way analysis of variance (ANOVA) was performed between the pre and post dataset in each group for the clinical data, followed by Sidak's multiple comparisons. For the dataset of score changes, an unpaired Student's *t*-test was performed between the *Placebo* and *CCFM1025* groups, with the Cohen's d to evaluate the effect size. The *P*-values for multiple comparisons were adjusted with family-wise significance, and 95% confidence interval and a $P < 0.05$ in all comparisons were considered statistical significance. Statistical analyses were performed using SPSS 22.0 and visualized using Prism 8.0.

3. Results

3.1. *B. breve CCFM1025 showed a better antidepressant-like effect than placebo*

The patients' HDRS-24 rating score significantly decreased from the baseline in both placebo- (pre: $M = 20.96$, $SD = 8.69$, post: $M = 13.92$, $SD = 8.09$; pre vs post: $F(1, 43) = 120.0$, $P < 0.001$) and CCFM1025-treated groups (pre: $M = 20.50$, $SD = 7.84$, post: $M = 8.90$, $SD = 6.40$; pre vs post: $F(1, 43) = 120.0$, $P < 0.001$; Fig. 2A. However, the score reductions in the two groups were statistically different. CCFM1025 decreased the score to a larger extend when compared with the placebo (placebo: $M = 6.44$, $SD = 5.44$; CCFM1025: $M = 10.40$, $SD = 6.85$; $t(43) = 2.163$, $P = 0.036$, $d = 0.640$; Fig. 2A. Similar results were observed in MADRS scores (placebo-pre: $M = 15.84$, $SD = 9.52$, placebo-post: $M = 10.92$, $SD = 8.15$; pre vs post: $F(1, 43) = 44.60$, $P = 0.003$; CCFM1025-pre: $M = 16.35$ $SD = 8.13$, CCFM1025-post: $M = 6.75$, $SD = 6.50$; pre vs post: $F(1, 43) = 44.60$, $P < 0.001$; Fig. 2B. CCFM1025 showed a better effect on reducing MADRS scores (placebo: $M = 4.92$, $SD = 7.15$; CCFM1025: $M = 9.60$, $SD = 7.37$; $t(43) = 2.152$, $P = 0.037$, $d = 0.645$; Fig. 2B. BPRS was used to comprehensively estimate the patients' psychiatric conditions, including somatization symptoms. The BPRS scores significantly decreased from the baseline in the CCFM1025 group (pre: $M = 31.80$, $SD = 5.79$, post: $M = 25.90$, $SD = 3.85$; pre vs post: $F(1, 43) = 26.03$, $P < 0.001$) but not the placebo group (pre: $M = 30.04$, $SD =$

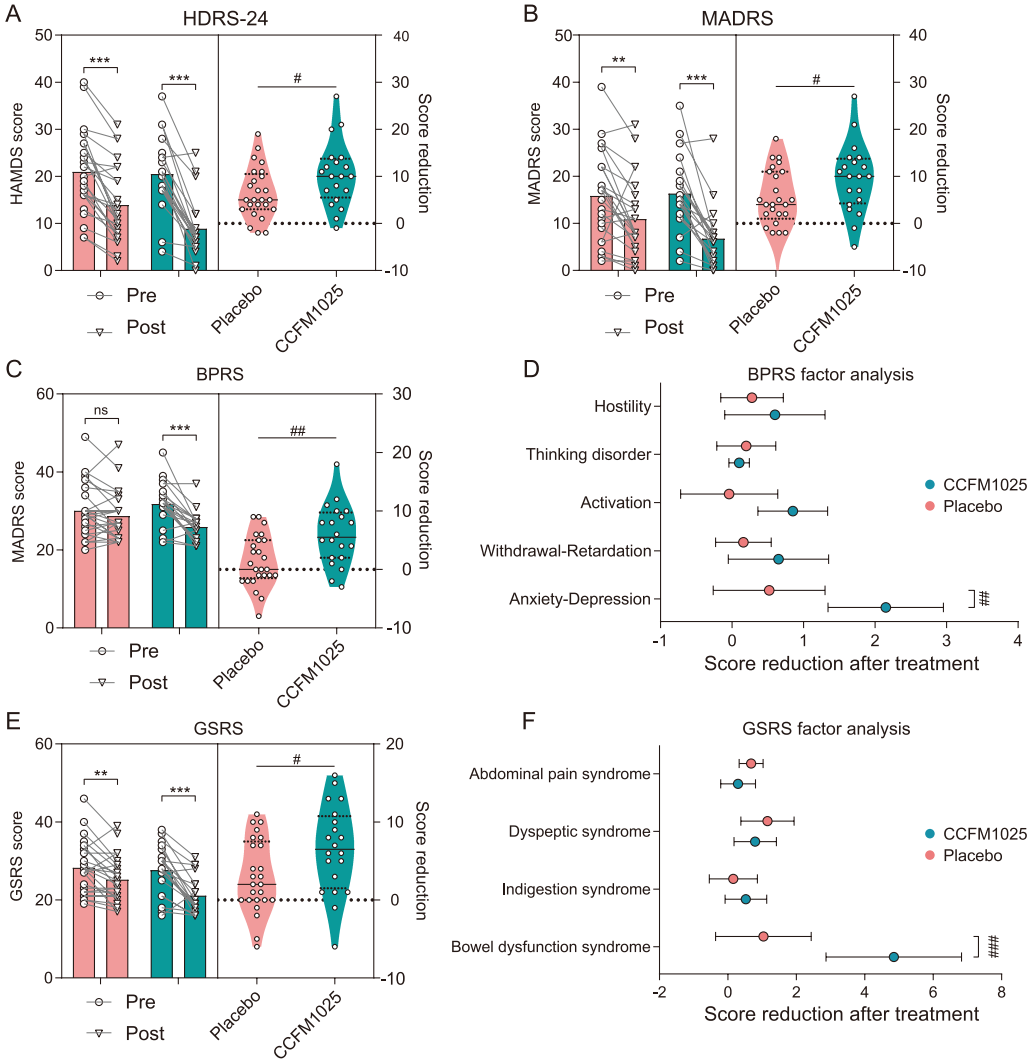

Fig. 2. *B. breve* CCFM1025's effect on alleviating depression symptoms. (A) The rating score change of the Hamilton Depression Rating scale-24 Items (HDRS-24). (B) The rating score changes of the Montgomery-Asberg Depression Rating Scale (MADRS. (C) The rating score changes of the Brief Psychiatric Rating Scale (BPRS). (D) Factor analysis of BPRS. (E) The rating score changes on the Gastrointestinal Symptom Rating Scale (GSRS). (F) Factor analysis of GSRS. In Figures A, B, C & E, a line connects each participant between day 0 (pre) and day 28 (post), and the reductions are represented as a histogram in the right panel. Colours distinguish groups. $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$ in Sidak's multiple comparisons test post two-way ANOVA. $^{\#}P < 0.05$, $^{\#\#}P < 0.01$, $^{\#\#\#}P < 0.001$ in the unpaired t-tests.

6.96, post: $M = 28.68$, $SD = 6.23$; pre vs post: $F(1, 43) = 26.03P = 0.2926$; Fig. 2C. The factor analysis of BPRS further confirmed that the participants' psychiatric problems were majorly attributed to the emotional disorders (anxiety and depression; Fig. 2D and CCFM1025 showed a better emotion-beneficial effect than the placebo ($t(43) = 2.982$, $P = 0.005$, $d = 0.899$; Fig. 2D. Gastrointestinal dysfunction is one of the most comorbid symptoms of depression, partly due to the intake of SSRIs (Steffens et al., 1997). The patients' GSRS rating scores decreased after both placebo (pre: $M = 28.24$, $SD = 6.85$, post: $M = 25.20$, $SD = 5.85$; pre vs post: $F(1, 43) = 38.75$, $P = 0.01$) and probiotic treatment (pre: $M = 27.65$, $SD = 6.43$, post: $M = 21.10$, $SD = 4.85$; pre vs post: $F(1, 43) = 38.75$, $P < 0.001$; Fig. 2E, but CCFM1025 showed a better effect (placebo: $M = 3.04$, $SD = 4.67$; CCFM1025: $M = 6.55$, $SD = 5.67$; $t(43) = 2.278$, $P = 0.027$, $d = 0.676$; Fig. 2E especially in attenuating bowel dysfunction syndrome ($t(43) = 3.357$, $P = 0.002$, $d = 0.994$; Fig. 2F.

3.2. The regulation on the serotonergic system may determine the CCFM1025's emotional and gastrointestinal effects

The CCFM1025's emotion- and gut-targeted effects suggested that the serotonergic system may participate in the process. We further measured the serum 5-HT and 5-HIAA levels (Fig. 3A&B). CCFM1025 significantly reduced the 5-HT turnover (placebo-pre: $M = 0.9$, $SD = 0.33$, placebo-post: $M = 0.91$, $SD = 0.47$; pre vs post: $F(1, 43) = 4.665$, $P = 0.996$; CCFM1025-pre: $M = 1.17$, $SD = 0.43$, CCFM1025-post: $M = 0.90$, $SD = 0.37$; pre vs post: $F(1, 43) = 4.665$, $P = 0.009$; Fig. 3C, and was more effective than the placebo (placebo: $M = -0.01$, $SD = 0.41$; CCFM1025: $M = 0.27$, $SD = 0.40$; $t(43) = 2.267$, $P = 0.029$, $d = 0.681$; Fig. 3D, which may be resulted from an elevated 5-HT ($F(1, 43) = 2.059$, $P = 0.823$; Fig. 3A and decreased 5-HIAA ($F(1, 43) = 5.405$, $P = 0.033$; Fig. 3B levels. The serum cortisol and inflammatory cytokines levels were measured. However, no statistical difference was observed in the concentration changes of serum cortisol (placebo: $M = 37.11$, $SD = 49.14$; CCFM1025: $M = 9.38$, $SD = 64.22$; $t(31) = 1.405$, $P = 0.169$, $d = 0.485$; Fig. 3E, serum TNF-α (placebo: $M = -2.547$, $SD = 16.36$; CCFM1025: $M = 1.880$, $SD = 25.31$; $t(40) = 0.684$, $P = 0.498$, $d = 0.208$; Fig. 3F, or serum IL-1β (placebo: $M = 1.049$, $SD = 6.514$; CCFM1025: $M = 0.3779$, $SD = 7.607$; $t(39) = 0.304$, $P = 0.763$, $d = 0.095$; Fig. 3G.

3.3. CCFM1025 showed a slight gut microbial modulation effect

Principal Component Analysis (PCA) visualized the gut microbial beta diversity of Aitchison distance (Fig. 4A. No significant difference was found among all groups (PERMANOVA results: F = 0095, $R^2 = 0.033$, $P = 0.684$). The Aitchison distance changes, previously described as volatility (Bastiaanssen et al., 2021), are compared between placebo and CCFM1025 groups, but no difference was found either ($t(42) = 0.361$, $P = 0.72$; Fig. 4A. For the alpha diversity, CCFM1025 increased the Chao 1 index ($F(1, 40) = 4.633$, $P = 0.035$) and observed operational taxonomic units (OTUs) ($F(1, 40) = 4.536$, $P = 0.062$), but not the Shannon index ($F(1, 39) = 0.102$, $P = 0.917$; Fig. 4B. Core microbiome analysis reveals the taxa detected in over 50% of the population exceeds a given abundance threshold. The results indicated that *Blautia*, *Bacteroides*, and *Faecalibacterium* are the top three core taxa in this cohort, and CCFM1025 showed no significant perturbation on the microbiome (Fig. 4C. Differential taxa (top 15) between pre and post were identified with random forest classification. In the CCFM1025-treated group, the abundance of *Desulfovibrio* ($t = 3.357$, $P = 0.005$) and *Faecalibaculum* ($t = 2.066$, $P = 0.047$) significantly increased (Fig. 4D. However, the placebo changed no taxa (Fig. 4D. The *Bifidobacterium* abundance was also analyzed to evaluate the adherence to probiotics. A significant OTUs count elevation was observed in the probiotics-treated group compared to the placebo group (Supplemental Fig.S3, which may partly result from the intake of CCFM1025.

3.4. CCFM1025 changed the tryptophan metabolism of the gut microbiome

Inspired by the significant change of peripheral 5-HT turnover, we investigated the tryptophan metabolism in the gut microbiome. A total of 14 tryptophan metabolites were determined. Eight were up-regulated by CCFM1025 but down-regulated by placebo, including tryptophan, 5-hydroxytryptophan (5-HTP), 5-HT, 5-HIAA, indole-3-acetamide, indole-3-lactic acid, and indole-3-propionic acid (Fig. 5A. Six metabolites showed significantly different concentration changes between the placebo and CCFM1025 groups, including tryptophan ($t(39) = 2.25$, $P = 0.030$), 5-HTP ($t(41) = 3.025$, $P = 0.004$), tryptamine ($t(36) = 3.76$, $P <$

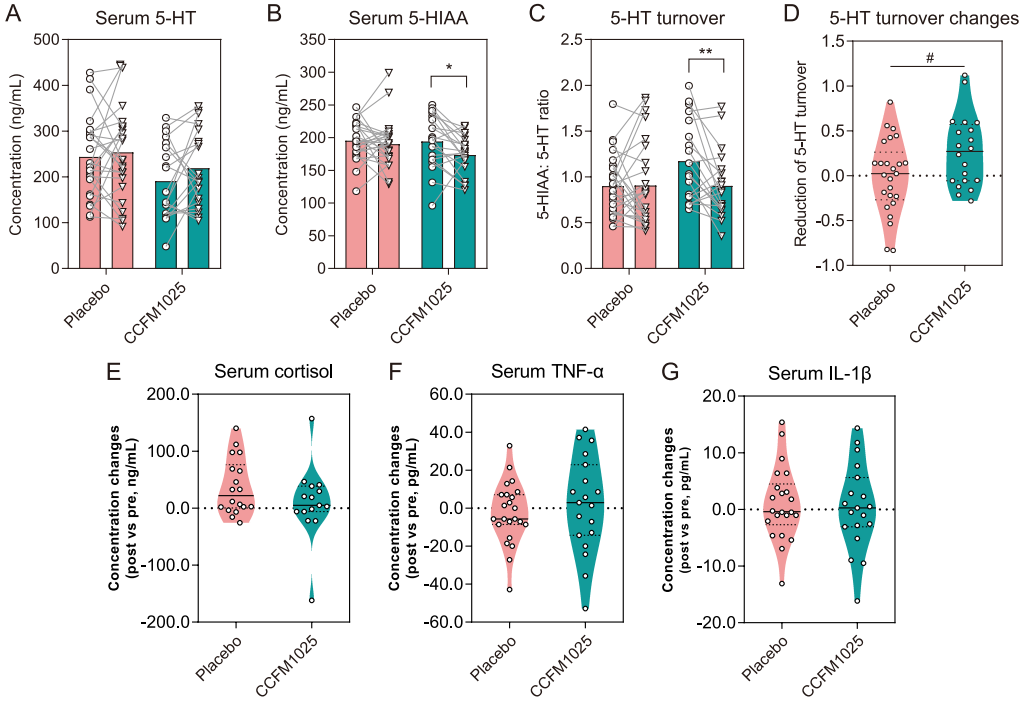

Fig. 3. *B. breve* CCFM1025's effect on neurotransmission and inflammation. (A) Serum 5-hydroxytryptamine (5-HT) levels. (B) Serum 5-hydroxyindoleacetic acid (5-HIAA) levels. (C) 5-HT turnover (the ratio of 5-HT: 5-HIAA). (D) The reduction of 5-HT turnover. (E) Serum cortisol concentration changes (post vs. pre) (F-G) Serum TNF-α and IL-1β concentration changes (post vs. pre). In Figures A-C, a line connects each participant between day 0 (pre) and day 28 (post), and the reductions of the concentration are represented as a histogram in the right panel. Colours distinguish groups. $^{*}P < 0.05$, $^{**}P < 0.01$ in Sidak's multiple comparisons test post two-way ANOVA. $^{\#}P < 0.05$ in the unpaired t-tests.

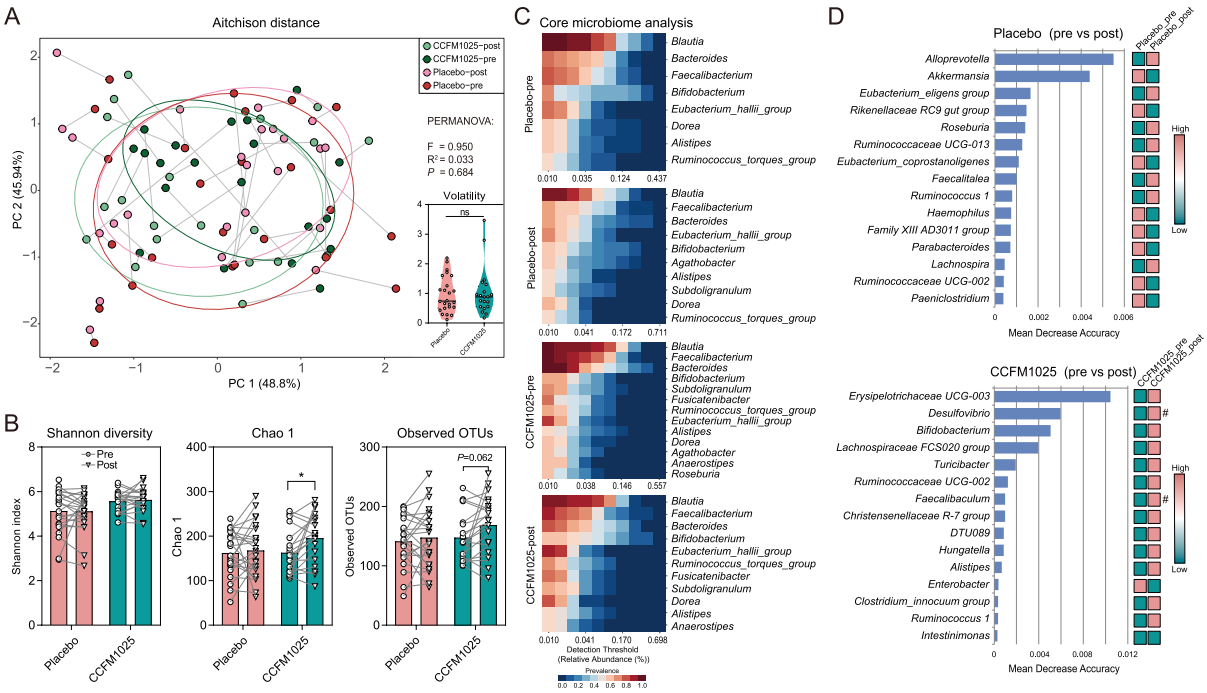

Fig. 4. *B. breve* CCFM1025′s effect on the gut microbiome. (A) Beta diversity. The structural similarity was estimated using principal component analysis (PCA) based on the Aitchison distance. The volatility is defined as the Aitchison distance change between pre and post. (B) Alpha diversity is characterized by Shannon, Chao 1, and Observed OTUs. $^{*}P < 0.05$ in Sidak's multiple comparisons test post two-way ANOVA. (C) Core microbiome analysis. The taxa shown are presented in over 50% of the population and a relative abundance of 0.01%. (D) The potential microbial biomarker was selected by random forest. The coloured boxes on the right indicate the relative abundance of each taxon. $^{\#}P < 0.05$ in the unpaired t-tests.

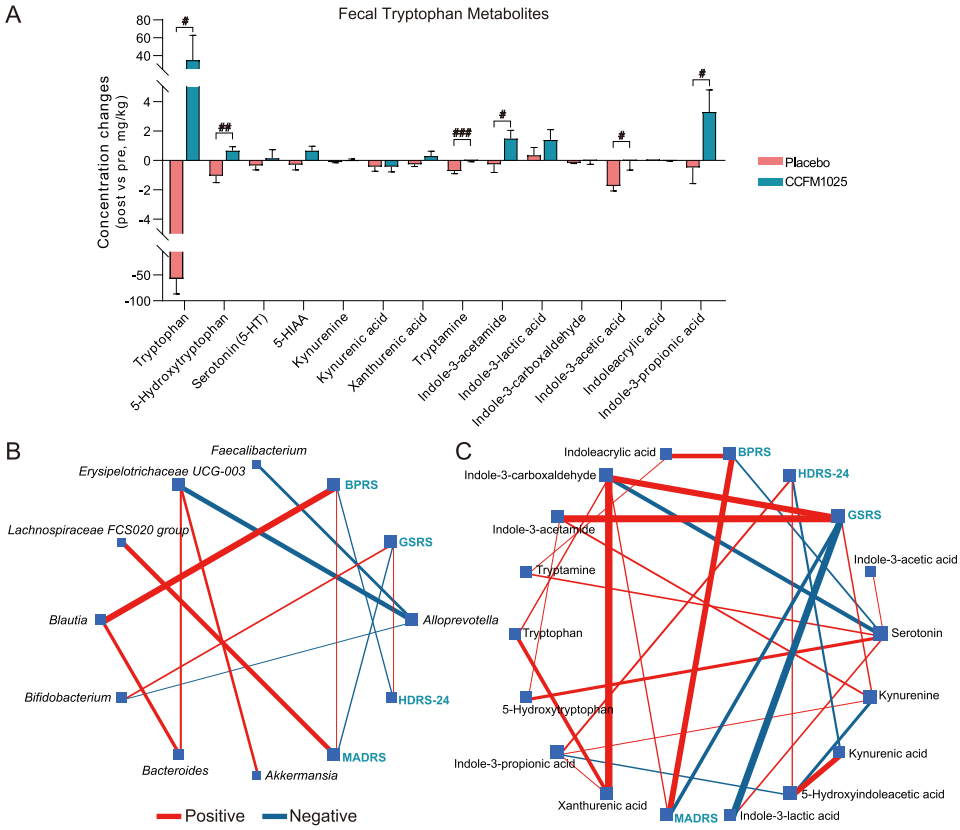

Fig. 5. *B. breve* CCFM1025′s effect on the gut tryptophan metabolism. (A) Concentration changes of the tryptophan metabolites in the CCFM1025 and placebo groups. $^{\#}P < 0.05$, $^{\#\#}P < 0.01$, $^{\#\#\#}P < 0.001$ in the unpaired t-tests. (B) Network of tryptophan metabolites and depression symptoms. The correlation was established between concentration changes of the tryptophan and psychometric and gastrointestinal rating scores. The size of each node is proportional to the number of connections. Red lines represent a positive correlation, and the blue lines represent a negative correlation. The width of the line represents the correlation strength. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

0.001), indole-3-acetamide ($t(40) = 2.199$, $P = 0.034$), indole-3-acetic acid ($t(40) = 2.506$, $P = 0.016$), and indole-3-propionic acid ($t(42) = 2.079$, $P = 0.044$; Fig. 5A. The correlation between gut microbiological changes and psychiatric symptoms changes was established by Debiased Sparse Partial Correlation (DSPC) network analysis (Basu et al., 2017). In the CCFM1025 group, *Blautia*, *Bifidobacterium,* and *Lachnospiraceae FCS020 group* are positively correlated with the psychometric and gastrointestinal improvements (Fig. 5B. Serotonin showed the highest degree (the number of connections it has to other nodes) of the network in the CCFM1025 group (Fig. 5C. Besides, indole-3-propionic acid and indole-3-acetamide, which were increased by CCFM1025, were positively correlated with the HDRS-24 (Coefficient = 0.092, $P = 0.01$) and GSRS (Coefficient = 0.351, $P < 0.001$) changes, respectively. However, in the placebo group, indole-3-acetamide was recognized as the dominant node of the network, which was negatively correlated with the BPRS changes (Coefficient = 0.042, $P = 0.039$, Supplemental Fig.S4.

4. Discussion

In this study, the psychotropic potential of *B. breve* CCFM1025, previously identified with antidepressant-like effect in mice, was further evaluated in MDD patients. As an adjunctive supplement, the intake of CCFM1025 significantly, and to a more considerable extent, reduced the patients' psychiatric and gastrointestinal symptoms compared with placebo. The effect may be correlated with the change of gut microbiome and tryptophan metabolism.

Modifying gut microbiota has become the most promising method among novel therapeutic strategies for mood disorders because of the increased understanding of the microbiome-gut-brain axis (Tremblay et al., 2020). Psychobiotics showed several unique advantages compared with traditional psychotropics. First, psychobiotics are a class of probiotics that has edibility and validated safety (Zagórska et al., 2020). Second, psychobiotics could directly communicate with the gut and microbiome, then produce positive effects on the gut through reshaping the internal micro-ecological balance (Cheng et al., 2021). Third, as the gut is the biggest digestive organ, its physiological changes could profoundly affect the whole organic system. Compared to traditional single-site-targeted drugs, psychobiotics address psychiatric problems by employing multiple physiological systems, including the immune system, enteric nervous system (ENS), and neuroendocrine system (Mörkl et al., 2020).

Specifically, CCFM1025 has been previously proven to promote the 5-hydroxytryptophan and serotonin (5-HT) synthesis of the gut enterochromaffin cells (Tian et al., 2020; Tian et al., 2019a). We found that CCFM1025 significantly decreased the patients' 5-HT turnover in the circulation. Unlike the results observed in animals, CCFM1025 did not cause a dramatic gut microbial perturbation in depressed patients, as the beta diversity and core microbes were not significantly changed. However, CCFM1025 and the placebo resulted in significantly different changes in the gut tryptophan level. As a kind of essential amino acid acquired from the diet, tryptophan is the only precursor for the biosynthesis of 5-HT *in vivo*, and over 90% of the 5-HT was synthesized in the gut (Alcaino et al., 2018). 5-HT is a well-studied neurotransmitter widely spread in the nervous system and is responsible for regulating multiple physiological processes, such as emotion, sleep, gut motility, and vasoconstriction (Chen et al., 2021; Mörkl et al., 2020).

Among all the mechanistic research trying to elucidate the interaction between the gut microbiome and brain functions, tryptophan metabolism is the connector that has received the most attention and recognition (Agus et al., 2018; Cryan et al., 2019). In recent years, accumulating evidence suggests that peripheral serotonin is related to emotional states. One of the most convincing aspects originated from the research on germ-free animals. Numerous studies have demonstrated that the presence or absence of gut microbiota influences anxiety- or depression-like behaviours (Dinan and Cryan, 2013; Neufeld et al., 2011) and is accompanied by neurochemical changes in the brain, such as the serotonin level and gene expression of second messenger pathways (Clarke et al., 2013; Heijtz et al., 2011; Zheng et al., 2016). Gut microbiota is proven to be essential in regulating the colonic serotonin biosynthesis and influencing the blood serotonin level, which results in downstream consequences for gastrointestinal motility, hemostasis, ossification, and brain functions (Bellono et al., 2017; Sugisawa et al., 2020; Yano et al., 2015). Both commensal host bacteria and orally administered exogenous strains can directly enhance the biosynthesis of serotonin from the enterochromaffin cells (which is the primary producer of the circulating serotonin), such as spore-forming bacteria, *Clostridium*, and *Bifidobacterium dentium* (Mandić et al., 2019; Turner et al., 2014; Yano et al., 2015). A recent study indicated that *Turicibacter* sanguinis can even re-uptake the intestinal serotonin, resulting in host lipid metabolism changes (Fung et al., 2019). Although the evidence comes from independent research groups, they establish a direct or indirect link between gut-derived serotonin and emotion states.

Gut microbiota is a complex ecosystem harbouring dense and diverse microbes and producing molecules that can trigger responses at local and distant levels, such as short-chain fatty acids and the secondary bile acids (Blacher et al., 2017). Unlike serotonin (5-hydroxytryptamine, 5-HT), peripheral tryptophan and 5-hydroxytryptophan (5-HTP, the precursor of 5-HT) can cross the BBB and participate in the biosynthesis of 5-HT in the brain, which is validated with the radioisotope [^{11}C]5-HTP and [^{11}C]methyltryptophan (a tryptophan analogue) tracking tests (Jacobsen, 2016; Visser et al., 2011). More importantly, oral administration of 5-HTP shows a clinically positive antidepressant-like effect, and adjunctive slow-released 5-HTP could powerfully and synergistically augment SSRI-induced serotonin elevation in the synaptic cleft (Jacobsen et al., 2016). It is worth noting that some novel metabolic biomarkers participating in the depression remission were identified, and most of them are indole derivatives, such as indole-3-acetamide, indole-3-acetic acid, and indole-3-propionic acid (Fig. 5A. Notably, the indole-3-propionic acid (IPA) changes are positively correlated with HDRS-24 score reduction (Fig. 5C. Indole and its derivatives are the direct products transformed from tryptophan by microorganisms (Agus et al., 2018). The majority of indole derivatives are ligands of aryl hydrocarbon receptor (AHR), which exerts potent effects on immune and metabolic function (Platten et al., 2019). Previously, we demonstrated that CCFM1025 could alleviate cortisol resistance and aseptic inflammation in the stress-induced depression mouse model (Tian et al., 2020). However, we found no significant changes in the basal cortisol and inflammatory markers in the patients' serum (Fig. 3E-G). Due to invasive sampling limitations, we did not get any evidence that might describe the IPA's intestinal immunological effect. More work is needed to figure out whether the IPA attenuates depression through regulating the immune system. These approaches provide evidence to support the peripheral tryptophan and its metabolite may causally affect the host's emotional state. However, undoubtedly, future investigations should focus on the dose and effect relationship between the gut-derived molecules and the host's neurobiological changes.

Several limitations need to be addressed in future work. First, the sample size needs to be expanded in the second round of human trials to generate more results, such as the sex difference in the treatment. Second, the patients' diet and medication were not strictly controlled except for the use of antibiotics. Since the gut microbiome was primarily affected by diet, the probiotics-induced gut microbial characteristics may be obscured. Strictly controlled diet and shotgun metagenomics should be applied in further studies to identify more specific findings. Third, the research staff reminded all subjects to take the probiotics every day with text messages; however, it was hard to supervise the subjects' actual probiotics adherence. A strain-specific primer should be developed to determine its abundance in the feces and calculate the actual adherence rate. Last but not least, since the CCFM1025 reduced the depression symptoms more significantly than placebo, whether CCFM1025 could speed up the recovery of the depression worth investigating in the following research. We believe resolving the issues above

would provide precise guidance for future probiotics application in managing psychiatric disorders.

5. Conclusion

In conformity with the predicted impact from preclinical research, *B. breve* CCFM1025 is a promising candidate psychobiotic strain that attenuates depression and associated gastrointestinal disorders. The mechanisms may be relevant to the modification of the gut microbiome and its tryptophan metabolism. Our findings generate some basic knowledge on using psychobiotics to improve emotional and physiological conditions. Further studies performed in larger scales with differential follow-up durations are warranted to establish a detailed assessment of CCFM1025′s therapeutic effects and to unravel the in-depth mechanisms underlying this effect.

6. Data availability statement

The data supporting the conclusions of this article are included in this published article and its supplementary files.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bbi.2021.11.023.

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